Advanced SQL

NULL FUNCTIONS

Exercise 2.1 Practical SQL Worksheet

Table 1: Employees

employee_id	name	department	salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

Q1: Show all employees with their salary. If salary is NULL, display 0.

SQL Query

```
SELECT employee_id,
       name,
    IFNULL(Salary,0) AS salary_with_default
 FROM Employees;
```

Output Results

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2: Show employee names with their department. If department is NULL, Show "Not Assigned".

Table 2: Orders

Order_id	Customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	202

Cancelled!!!

Q2: Show employees name with their department. If department is NULL, show "Not assigned.

SQL Query

```
SELECT employee_id,
       name,
    IFNULL(department, 'Not Assigned') AS department_name
 FROM Employees;
```

Output Results

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dona	Finance

Table 2: Orders

Order_id	Customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

Q3: Find orders with NULL customer_id using ISNULL().

SQL Query

```
SELECT order_id,
       Customer_id
  FROM Orders
  WHERE customer_id IS NULL;
```

Output Results

Order_id	Customer_id
103	NULL

Q4: Show all orders. If delivery_date is NULL, show 'Pending'.

SQL Query

```
SELECT  order_id,
        customer_id,
     IFNULL (delivery_date, 'Pending') AS delivery_status
  FROM Orders;
```

Output Results

Order_id	Customer_id	delivery_Status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: Students

Student_Id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

Q5: Show all Students and their grades. Replace NULL with 0.

SQL Query

```
SELECT  student_id,
        name,
     IFNULL (grade,0) AS final_grade
  FROM Students;
```

Output Results

Student_Id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6: Count students who havent been graded.

SQL Query

```
SELECT COUNT(*) AS not_graded_count
   FROM Students
  WHERE grade IS NULL;
```

Output Results

not_graded_count
1

Table 4: Products

Product_Id	Name	Price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

Q7: Show product name, Price, and final after discount (assume 0 if discount is NULL)

Q7: SQL Query

```
SELECT product_id,
       name,
       price - IFNULL(discount,0) AS final_price
  FROM Products
```

Output Results

Product_id	name	Final_Price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Q8: Count how many customers have no email.

Table 5: Customers

Customer_id	name	email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Nia	NULL

SQL Query

```
SELECT COUNT(*) AS missing_email_count
  FROM Customers
WHERE email IS NULL;
```

Output Results:

Missing_email_count
2

Q9: Show all customers with email. If NULL, display "NO Email".

SQL Query

```
SELECT customer_id,
       name,
   IFNULL(email,'NO Email') AS email_display
```

Output Results

customer_id	name	email_display
1	Linda	NO Email
2	Joseph	joseph@mail.com
3	Mia	NO Email

Table6: Payments

payment_id	Method	Status
301	Credit	NULL
302	PayPal	Success
303	NULL	Failed

Q10: Show payment details with method replaced by "Unknown" if NULL.

SQL Query

```
SELECT payment_id,
   IFNULL(Method,'Unknown') AS method_display,
   Status
FROM Payments;
```

Output Results

payment_id	method_display	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Table 7: Inventory

item_id	item_name	quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

Q11: Show Items and their quantity (0 if NULL).

SQL Query

```
SELECT Item_id,
       Item_name,
    IFNULL(quantity, 0) AS quantity_Checked
  FROM Inventory;
```

Output Results

Item_id	Item_name	quantity_Checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employee_Extra

emp_id	bonus	Commission
1	NULL	300
2	100	NULL
3	NULL	NULL

Q.12. Show employee ID and the first available value among bonus or commission.

SQL Query

```
SELECT emp-id,
    COALESCE (bonus, Commission) AS first_available_reward
 FROM Employees_Extra;
```

Output Results

emp_id	first_available_reward
1	300
2	100
3	NULL

Table 9: Classes

class_id	Subject	room
11	Math	NULL
12	Science	Lab A
13	English	NULL

Q.13. Count classes that dont have a room assigned

SQL Query

```
SELECT COUNT(*) AS no_room_count
    FROM Classes
    WHERE room IS NULL;
```

Output Results

no_room_count
2

Table 10: Attendance

student_id	date	Status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

Q14. Show attendance records with status. Replace NULL with "Not Marked".

SQL Query

```
SELECT student_id,
       date,
   IFNULL(status,'Not Marked') AS attendance_status
  FROM Attendance;
```

Output Results

Student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11: Bank_Accounts

account_id	account_type	balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

Q15. Show account ID, account_type (or 'Unknown'), and balance (or 0).

SQL Query

```
SELECT account_id,
    IFNULL(account_type, 'Unknown') AS type_display,
    IFNULL(balance, 0) AS balance_checked
  FROM Bank_Accounts;
```

Output Results

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12: Projects

project_id	title	start_date	end_date
1	Website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

Q16. Show all projects with a start date. If start_date is NULL, display 'TBD'.

SQL Query

```
SELECT project_id,
       title,
    IFNULL(start_date, 'TBD') AS start_display
 FROM Projects;
```

Output Results

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13: Reviews

review_id	product_id	Comment	rating
1	501	Great Product	4
2	502	NULL	NULL
3	503	Works fine	3

Q 17. Display reviews showing comment (or 'NO Comment') and rating (or 0)

SQL Query

```
SELECT review_id,
       product_id,
    IFNULL(comment, 'NO comment') AS comment_display,
    IFNULL(rating, 0) AS rating_display
 FROM Reviews;
```

Output Results

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	NO Comment	0
3	503	Works fine	3

Table 14: Suppliers

Supplier_id	name	phone	alt_phone
1	Global Goods	NULL	123456789
2	Best Supplies	987654321	NULL
3	ValueSource	NULL	NULL

Q18. Show the supplier contact number. Use COALESCE (phone, alt_phone, 'NO Contant').

SQL Query

```
SELECT supplier_id,
       name,
  COALESCE (phone, alt_phone, 'NO Contact') AS Contant_number
  FROM Suppliers;
```

Output Results

Supplier_id	name	Contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	NO Contact

Table 15: User_Settings

user_id	theme	language	timezone
1	NULL	English	NULL
2	Dark	NULL	UTC+1
3	NULL	NULL	NULL

Q19. Show all users and their preferences. Replace all NULLs with defaults:

- Theme -> "Light"
- Language -> "English"
- Timezone -> "UTC"

SQL Query

```
SELECT user_id,
       IFNULL(theme,'Light') AS theme_set,
       IFNULL(language'English') AS language_set
       IFNULL(timezone,'UTC') AS timezone_set
  FROM User_Settings;
```

Output Results

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+
3	Light	English	UTC

Table 16: Maintenance

record_id	machine_id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

Q20. Show maintenance log with

- issue - default to "Unknown Issue"
- technician - default to "Not Assigned"

SQL Query

```
SELECT record_id,
       machine_id,
       IFNULL (issue, 'Unknown Issue') AS issue_log,
       IFNULL (technician, 'Not Assigned') AS technician_name
FROM Maintenance;
```

Output Results

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex